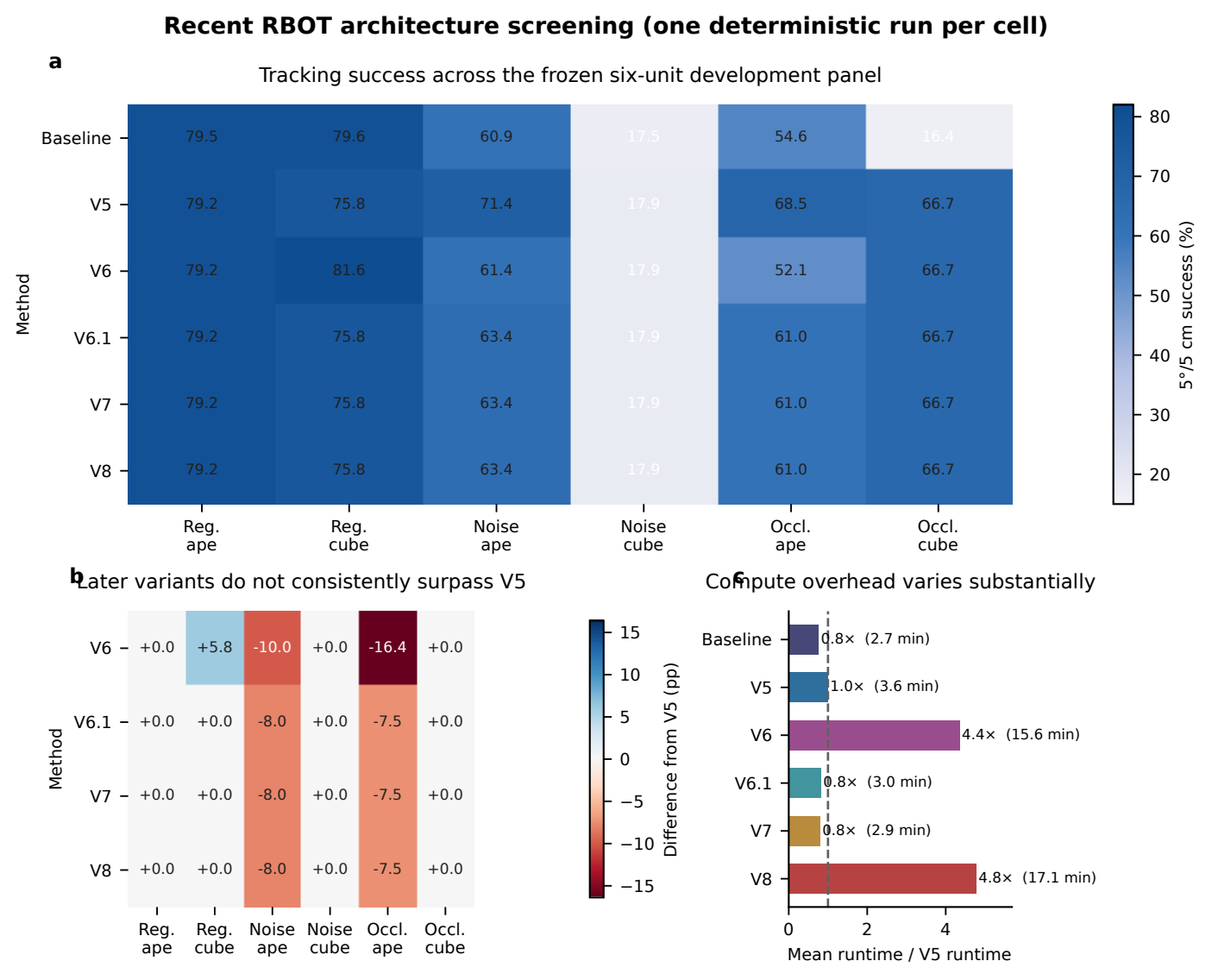
Recent RBOT architecture screening (one deterministic run per cell)
a
Tracking success across the frozen six-unit development panel
Method
Baseline
V5
V6
V6.1
V7
V8
79.5 79.6 60.9 17.5 54.6 16.4
79.2 75.8 71.4 17.9 68.5 66.7
79.2 81.6 61.4 17.9 52.1 66.7
79.2 75.8 63.4 17.9 61.0 66.7
79.2 75.8 63.4 17.9 61.0 66.7
79.2 75.8 63.4 17.9 61.0 66.7
Reg. ape
Reg. cube
Noise ape
Noise cube
Occl. ape
Occl. cube
5°/5 cm success (%)
80
70
60
50
40
30
20
b
Later variants do not consistently surpass V5
Method
V6 +0.0 +5.8 -10.0 +0.0 -16.4 +0.0
V6.1 +0.0 +0.0 -8.0 +0.0 -7.5 +0.0
V7 +0.0 +0.0 -8.0 +0.0 -7.5 +0.0
V8 +0.0 +0.0 -8.0 +0.0 -7.5 +0.0
Reg. ape
Reg. cube
Noise ape
Noise cube
Occl. ape
Occl. cube
15
10
5
0
−5
−10
−15
Difference from V5 (pp)
c
Compute overhead varies substantially
Baseline 0.8× (2.7 min)
V5 1.0× (3.6 min)
V6 4.4× (15.6 min)
V6.1 0.8× (3.0 min)
V7 0.8× (2.9 min)
V8 4.8× (17.1 min)
0
2
4
Mean runtime / V5 runtime